

2) Hull & Hull Outfit Part

1. Hull Part

C O N T E N T S

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2)	Progressive Speed Trial	2~5
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● NOTE

As this vessel(N 208, "DUBAI ANGEL") is the subsequent sister vessel of "DUBAI PRINCESS" (Hull No. N 198) the following tests were not carried out during this sea trial.

Therefore, the results for this vessel were obtained from the sister vessel.

- Turning Circle test
- Crash Stop Astern test at half ahead speed
- Z-Maneuvering test

[illegible]

HHIC	Test No. H-	PAGE	1
	Sea Trial Results for 1) Draft Measurement	Ship No.	N208
		Date	10. 09. 03

Condition : Sea Trial Condition	Measured Time AM 09.30
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Item		Method	Measured Draft					
			Aft	Midship	Fore			
(1)	Draft at Draft Marks	Port Side	13.670	13.620	13.660			
		Starboard Side	13.670	13.630	13.660			
(2)		Mean	13.670	13.625	13.660			
(3)	Apparent Trim	Fore mark - Aft mark	-0.010					
(4)	Draft Correction Factor		-0.0746	-0.0032	0.0394			
(5)	Draft Mark Correction	(3) x (4)	0.001	0.000	0.000			
(6)	Draft at (AP)(MIDSHIP)(FP)	(2) + (5)	13.671	13.625	13.660			
(7)	Factor		1	6	1			
(8)	Deflection Correction	(6) X (7)	13.671	81.750	13.660			
(9)	Mean Draft	Sum (8) / 8	13.635					
(10)	Actual Trim	Draft at (FP) - (AP)	-0.011					
(11)	Displacement & LCB(LCG) from the Trimmed Hydrostatic Table							
		Item	Displacement		LCB (=LCG)			
		Trim(m) Draft(m)	-0.500	-0.011	0.000	-0.500	-0.011	0.000
		13.600	121235.1	121215.4	121215.0	7.385	8.063	8.078
		13.700	122217.8	122196.5	122196.0	7.317	7.990	8.005
		13.635	121558.8 ton			8.037 m		
(12)	Actual S.G. of Sea Water	(As measured)		1.01900				
(13)	Displacement Correction due to S.G.	(11) x {(12) - 1.025/ 1.025}		-711.6				
(14)	Actual Disp't after All Corrections	(11) + (13)		120847.2				



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Sea Trial Results for
2) Progressive Speed Trial

Ship No.

N208

Date

10. 09 .04

Trial date : 2010.09.04

Trial Place : Off Busan

Ship Condition : Trial Ballast Condition

Draft at FP 13.660 m

Trim by Stern -0.011

Draft at AP 13.671 m

Displacement 120847.2

Draft, mean 13.635 m

Propeller Immersion 184%

Weather & Sea Condition :

Weather : Fine

Sea Condition : Calm

Depth of Sea Water : 128

Temperature of Atmosphere 28

Temperature of Sea Water : 26

Specific Gravity of Sea Water 1.019

Trial Result

M/E Load	Run No.	Course of Run	Time in	Speed (knot)	M/E rpm	M/E BHP(kW)	Wind		Wave (Average)	
							Vel.(m/s)	dir.(deg)	hgt.(m)	dir.(deg)
Abt. 90%	1	200	13:07	15.146	106.4	12565.0	9.5	S 25.5	1.0	60
	2	20	14:16	15.929	106.4	12533.0	5.8	P 23.2	1.0	60
	Mean			15.538	106.4	12549.0				
Abt. 100%	1	200	15:37	15.121	109.3	13911.0	9.8	S 22.5	1.0	60
	2	20	16:45	16.831	109.8	14068.0	7.4	P 25.6	1.0	60
	Mean			15.976	109.6	13989.5				

* The M/E power used in this report was measured by HHIC's Torsion-meter.

SPEED TRIAL ANALYSIS

BY NSMB METHOD

SHIP NAME: N208 EMARAT TRIAL DATE: 2010.09.03
 * LBP = 239.000 M * BREADTH = 44.000 M * TA/TF = 13.671/13.660 M
 * DISPL = 120847. TONS * T MEAN = 13.665 M

SPEED TRIAL RESULTS

LOAD	RUNNING	WIND	WIND	SHIP	RATE	SHAFT	CURRENT	INNING
(%)	DIRECT.	DIR.	SPEED	SPEED	REVS.	POWER	SPEED	TIME
(%)	(DEG)	(DEG)	(M/S)	(KTS)	(RPM)	(KW)	(KTS)	
90.	200.	S 25.5	9.5	15.146	106.40	12565.	0.000	13: 7
90.	20.	P 23.2	5.8	15.929	106.40	12533.	0.000	14:17
				15.538	106.40	12549.		
100.	200.	S 22.5	9.8	15.121	109.30	13911.	0.000	15:37
100.	20.	P 25.6	7.4	16.831	109.80	14068.	0.000	16:45
				15.976	109.55	13990.		

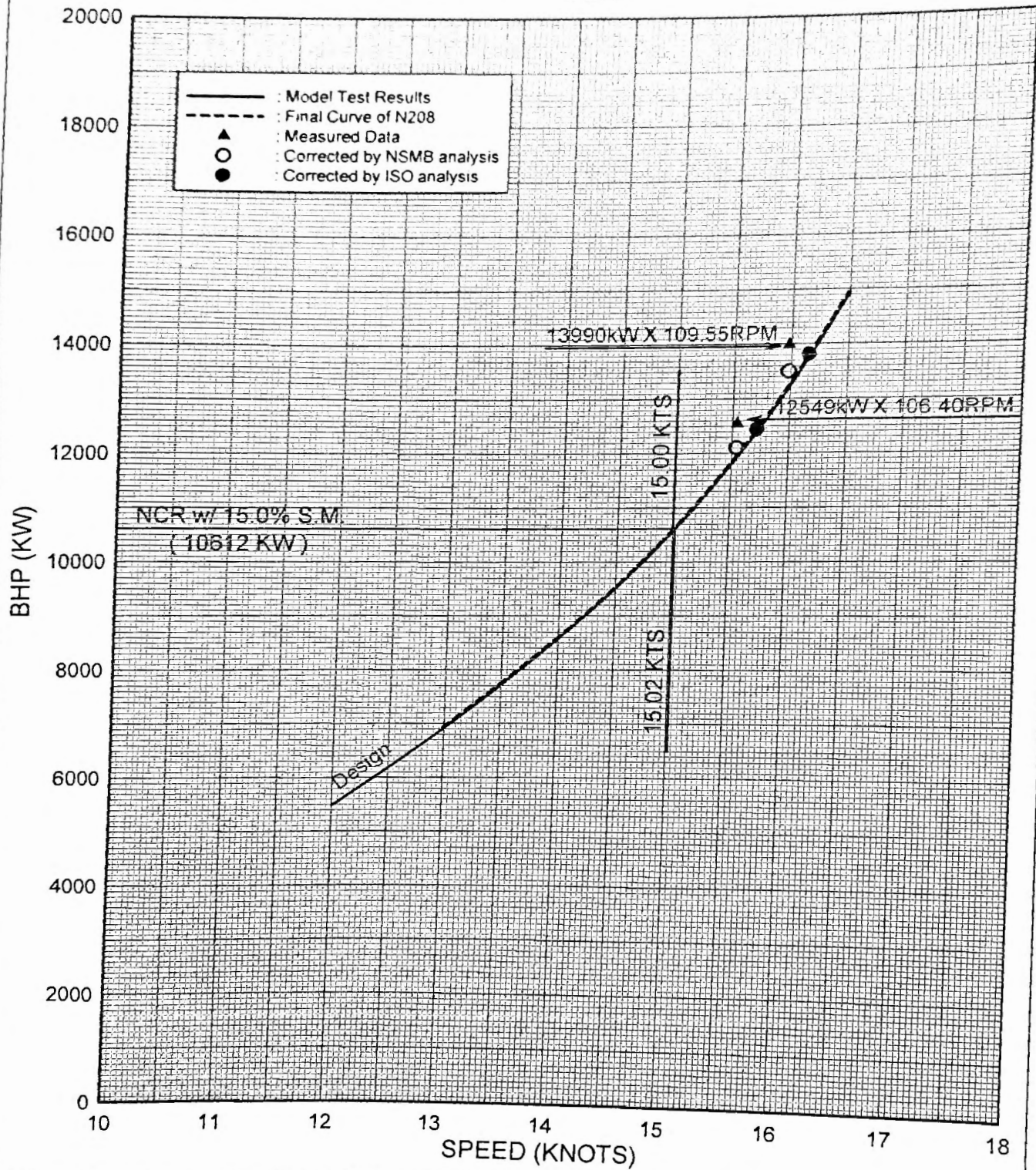
CORRECTION TO BEAUFORT NO. 0 (CALM SEA)

LOAD	RUNNING	TRUE	TRUE	WIND	CORR.	SHIP	CORR.	CORR.	DISPL.
	DIRECT.	W.DIR	W.S	SCALE	FACTOR	SPEED	RPM	BHP	CORR.
(%)	(DEG)	(DEG)	(KTS)	(B.N)		(KTS)		(KW)	RPM
90.	200.	279.2	8.09	3.4	1.036	15.146	105.16	12129.	105.06
90.	20.	238.6	7.12	3.2	1.032	15.929	105.30	12150.	105.21
				3.3	1.034	15.538	105.23	12139.	105.13
100.	200.	271.2	7.70	3.3	1.034	15.121	108.08	13451.	107.98
100.	20.	258.2	7.32	3.2	1.032	16.831	108.64	13626.	108.54
				3.3	1.033	15.976	108.36	13539.	108.26

* DISPLACEMENT CORRECTION : BY ADMIRALTY COEFFICIENT
 MODEL TEST DISPLACEMENT :120343. TONS

SPEED-POWER CURVE

N-208

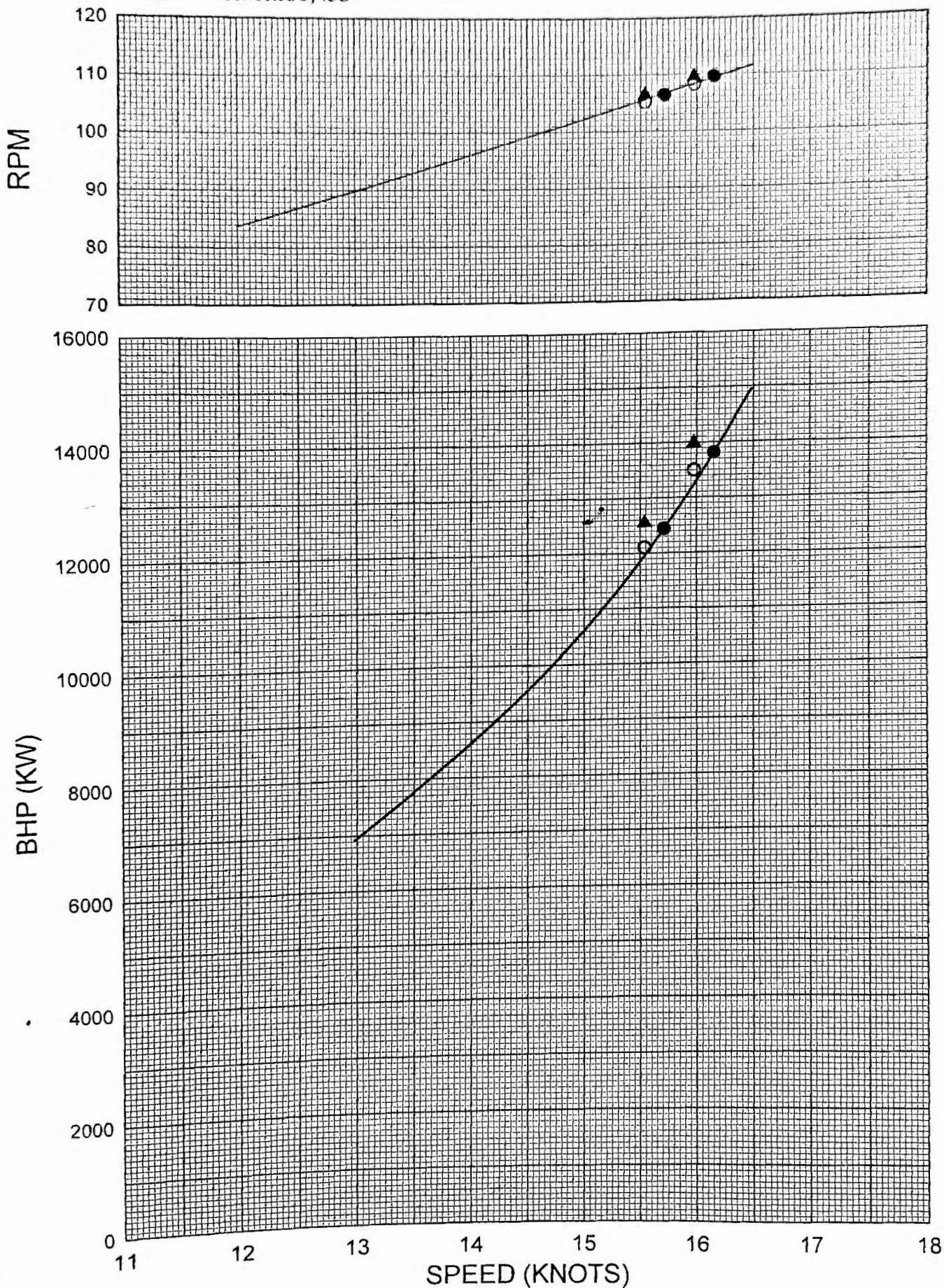


DESIGN LOAD

Service Speed at NCR w/ 15% S.M. : 15.00 KTS

SPEED-POWER CURVE

- : Final Curve of N208
- ▲ : Measured on trial
- : Corrected by NSMB
- : Corrected by ISO

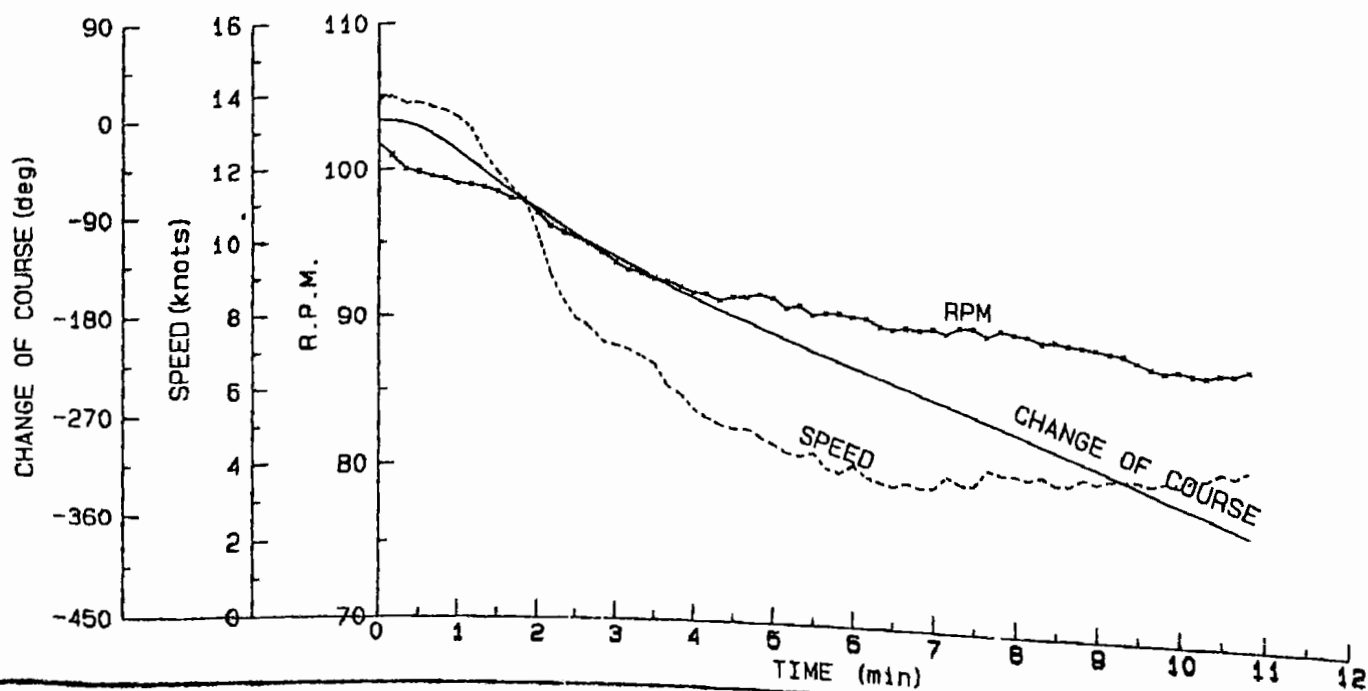
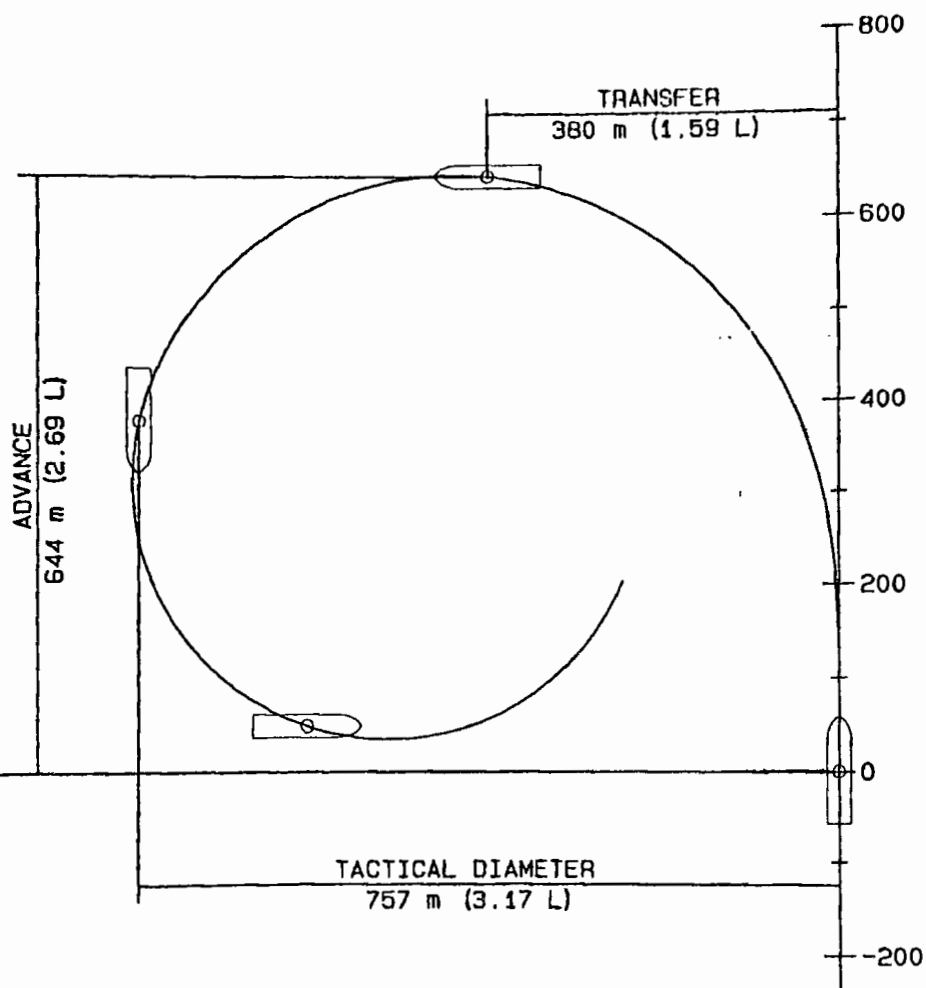




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	Sea Trial Results for 3) Turning Circle Test		Ship No.	N198
			Date	09.01.12
Trial date : 2009. 01. 12		Trial Place : Off Busan		
Ship Condition : Trial Ballast Condition		Sea Condition		
Draft, Fore	13.589 m	Weather	Windy	
Draft, Aft	13.756 m	Sea Condition	Rough	
Displacement	122069.3 ton	Wind Dir. & Velocity	S60°, 9.0 m/s	
		Depth of Sea Water	121 m	
Trial Result at Full Ahead Speed (Ful Speed)				
		Port Turn	Starboard Turn	
Clock Time		19:35	19: 00	
Base Course		200°	200°	
Rudder Angle		S - 60°	S - 55°	
Shaft rpm		100 rpm	100 rpm	
Ship speed		13.9 knots	14.0 knots	
Turning Time To (sec)	090 deg	138	124	
	180 deg	282	278	
	270 deg	471	465	
	360 deg	665	659	
	450 deg	855	860	
	540 deg	1045	1078	
	630 deg	1245	1277	
	720 deg	1440	1480	

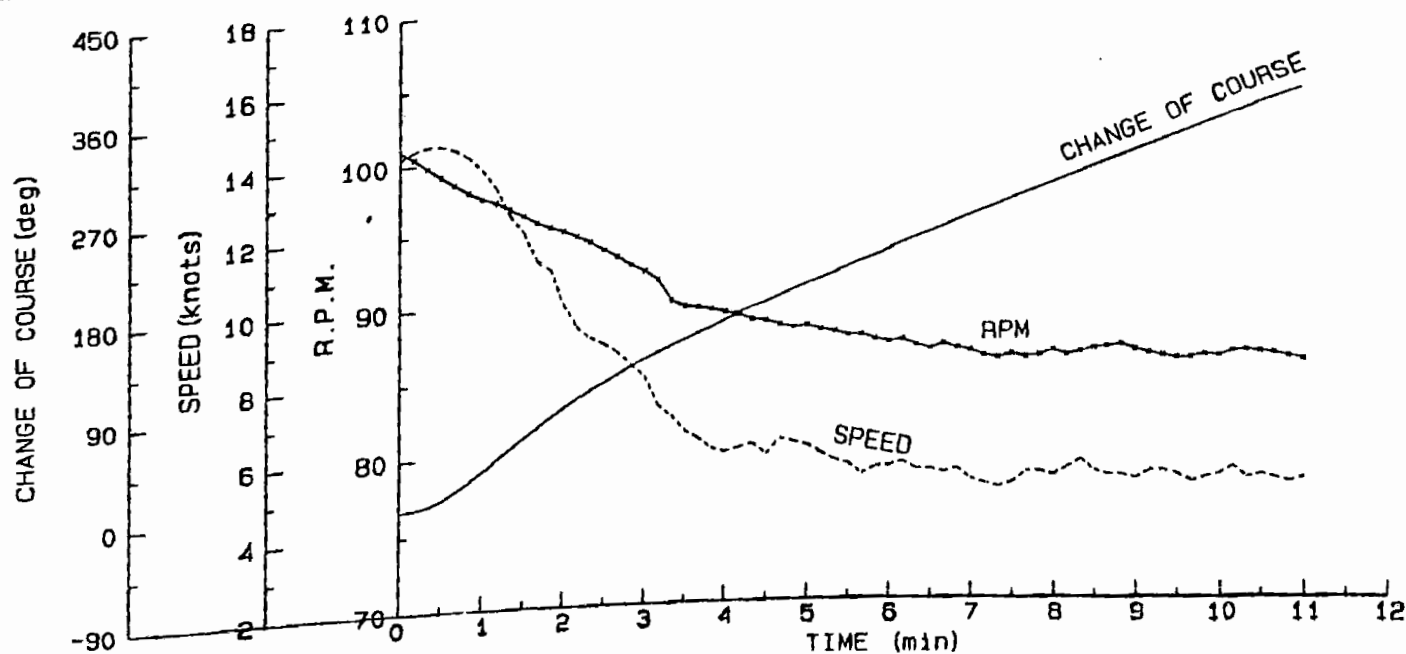
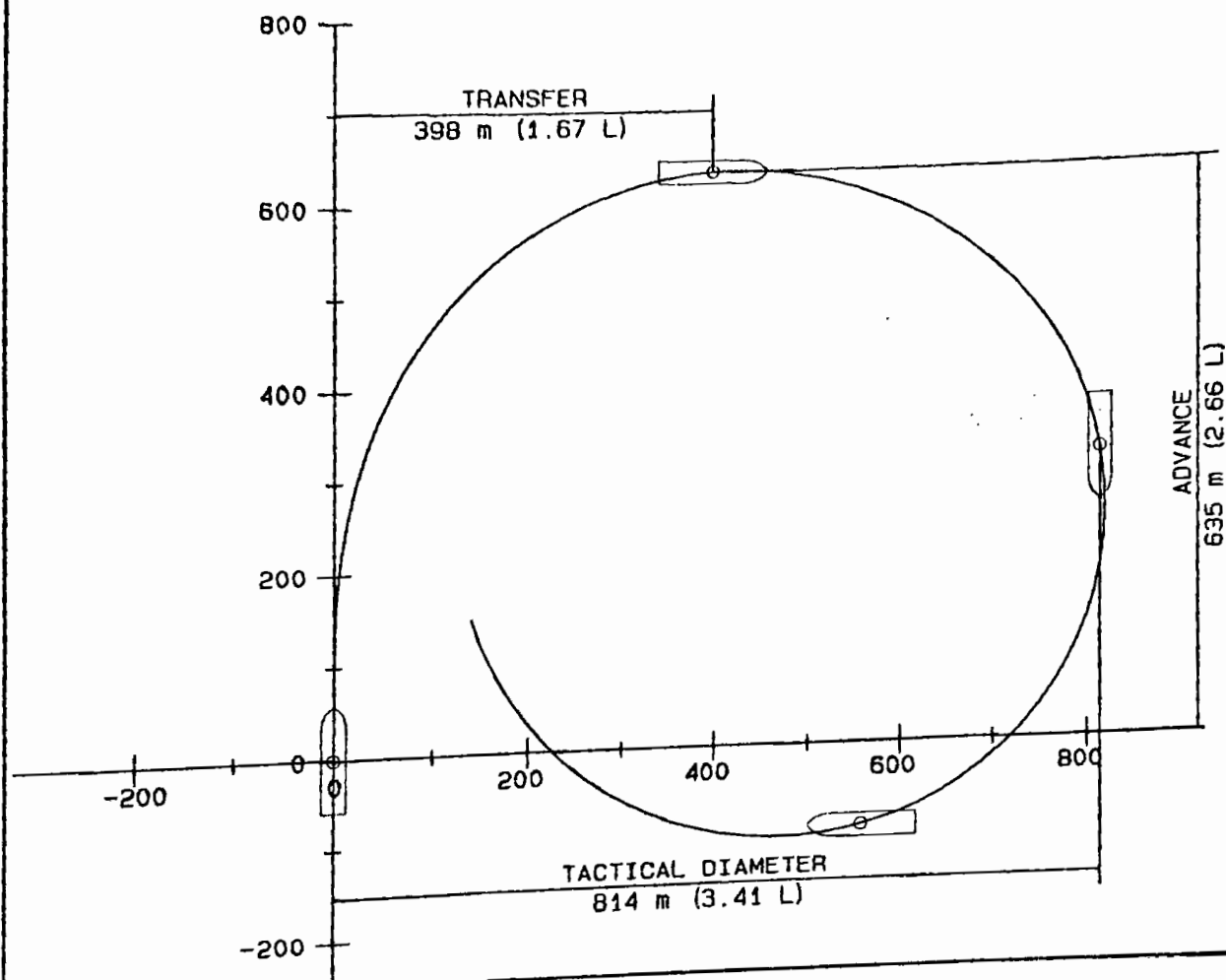
TURNING CIRCLE TO PORT

(BALLASTED CONDITION at IMO SPEED)



TURNING CIRCLE TO STARBOARD

(BALLASTED CONDITION at IMO SPEED)





Test No. H-

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Sea Trial Results for
4) Crash Stop Astern Test

Ship No.

N208

Date

10.09.05

Crash Stop Astern Test

Trial date : 2010. 09. 05

Trial Place : OFF PUSAN

Ship Condition : Trial Ballast Condition

Sea Condition :

Draft, Fore

13.660 m

Weather

Fine

Draft, Aft

13.671 m

Sea Condition

Calm

Displacement

120847.2

Wind Dir. & Velocity

S-10°, 12.9 m/s

Depth of Sea Water

129

Trial Result

IMO speed to Full Astern

Clock Time

00 : 25

Base Course

1.9

Shaft rpm

105

Ship speed

16.2 kts

Time to Start Shaft Astern

6min. 02sec.

Time to Stop Ship

12 min. 54sec.

Time to Reach Astern Steady rpm

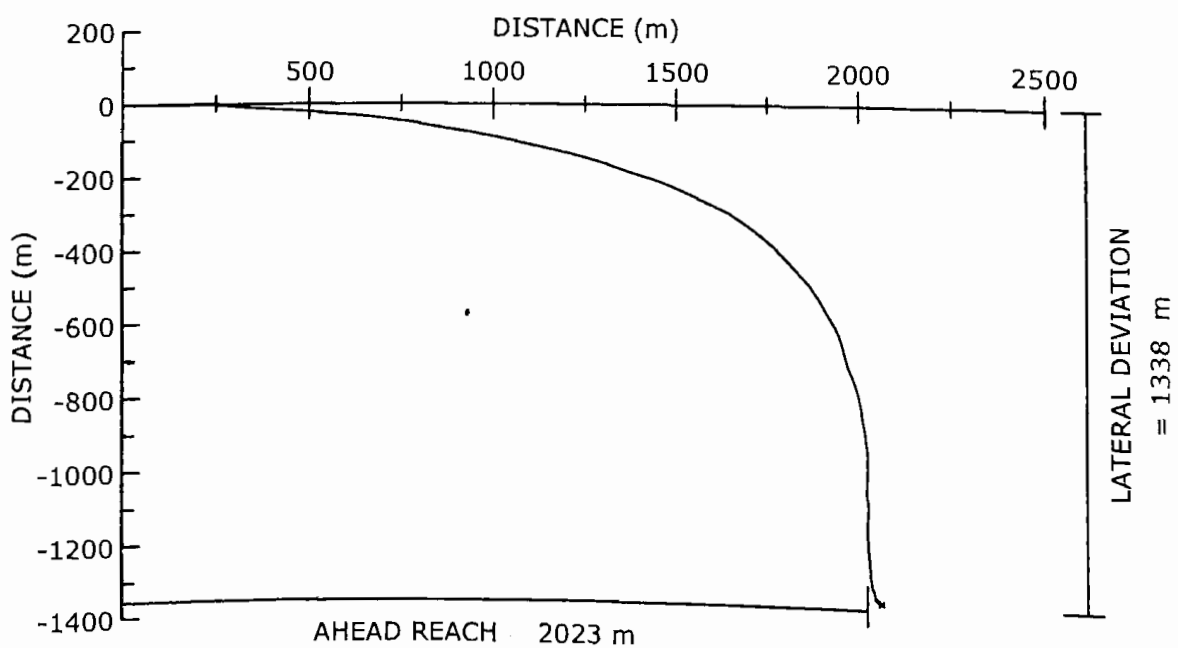
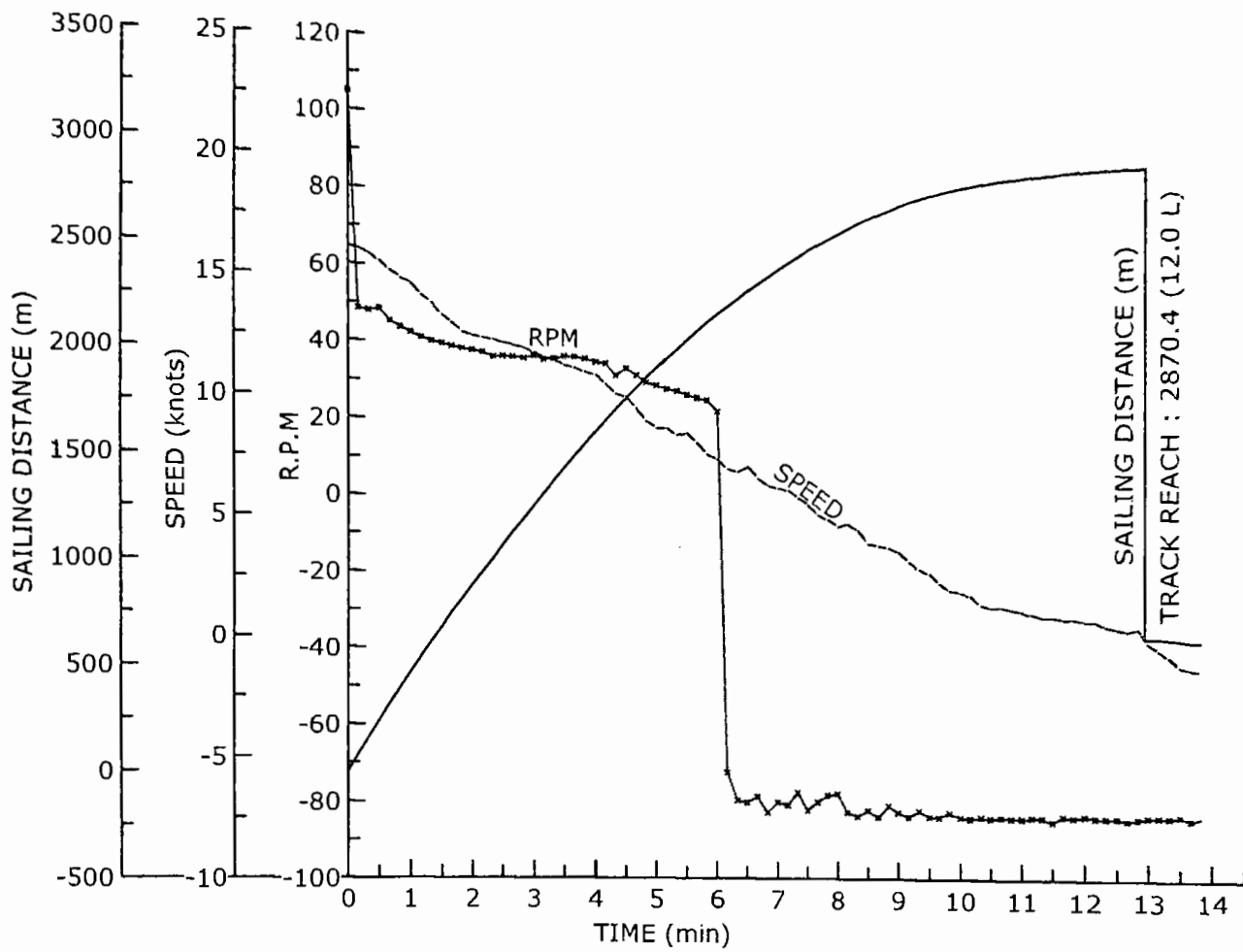
10 min. 09sec.

Astern Steady rpm

-84

CRASH STOP ASTERN

DESIGN LOADED CONDITION at FULL SPEED

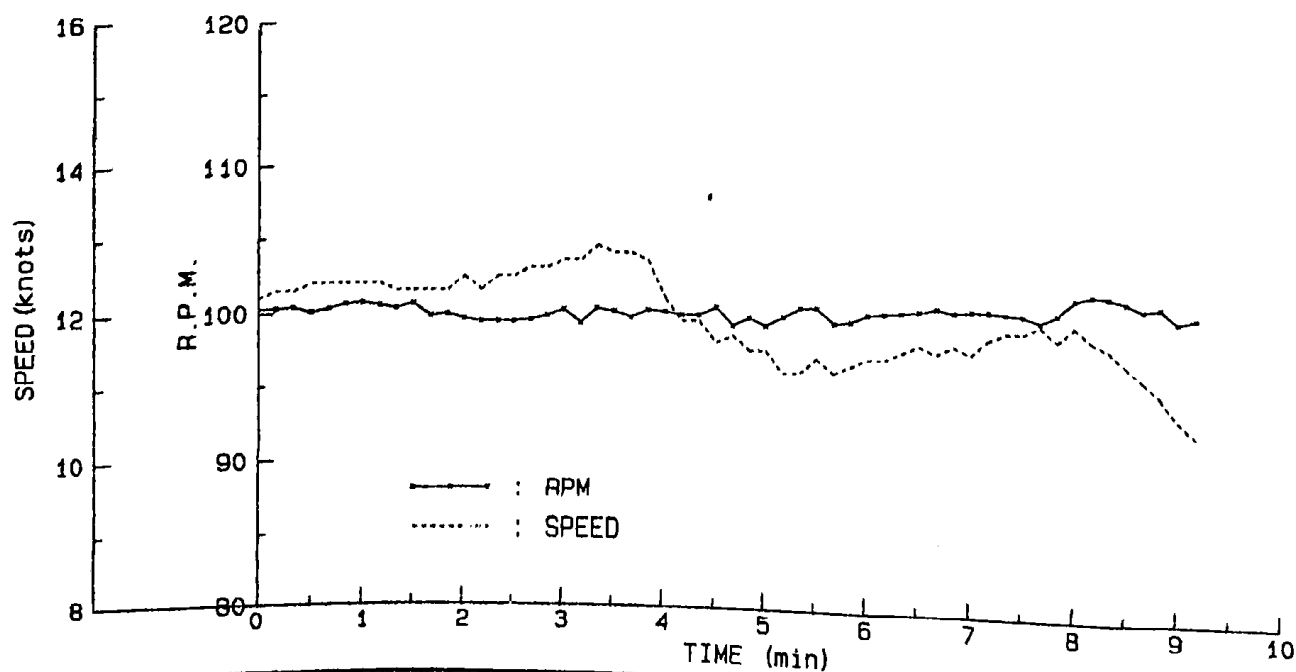
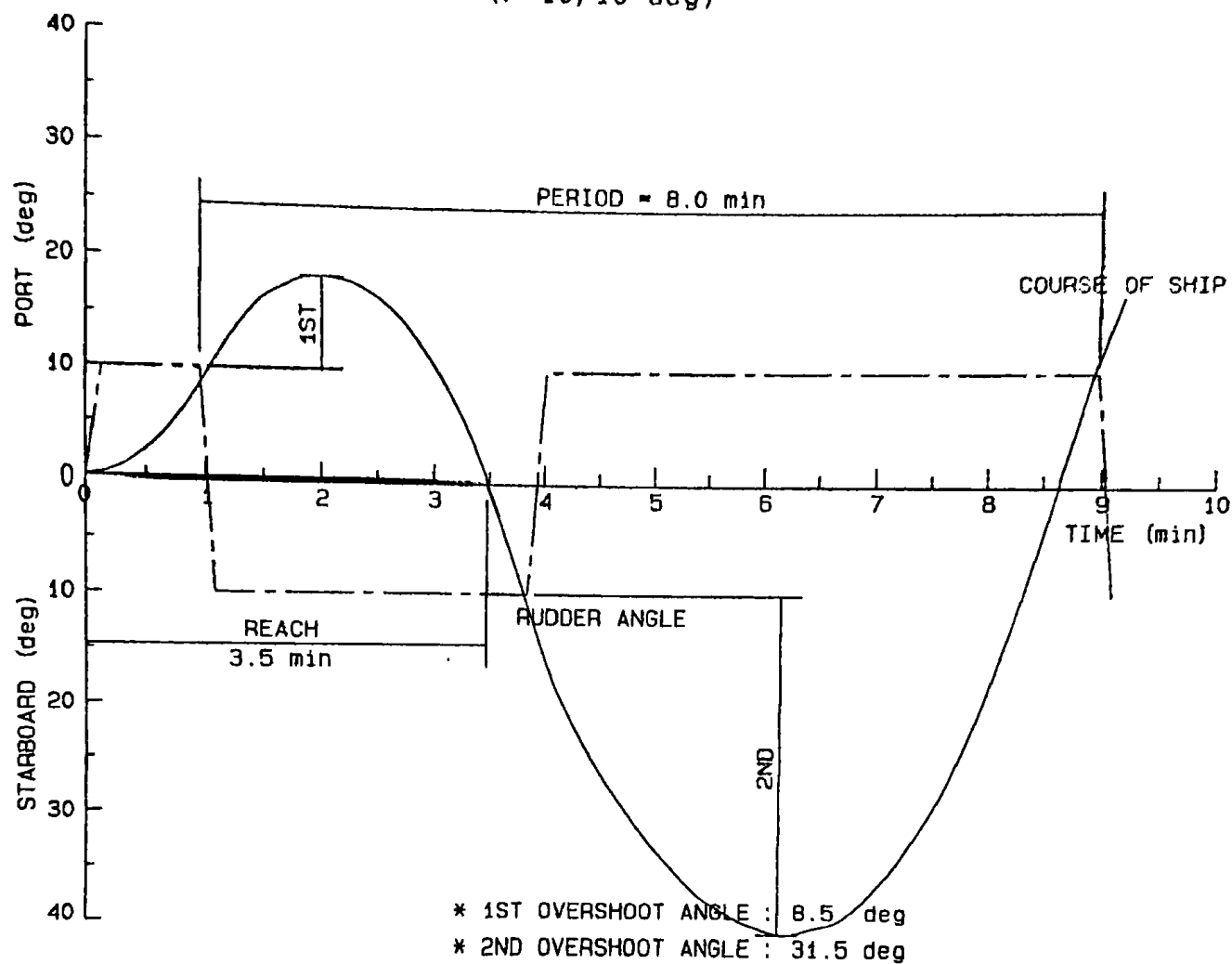


			Page	11
	Sea Trial Results for 5) "Z" Maneuver Test		Ship No.	N198
			Date	09.01.12
Trial date : 2009. 01. 12		Trial Place : Off Busan		
Ship Condition : Trial Ballast Condition		Sea Condition		
Draft, Fore	13.589 m	Weather	Windy	
Draft, Aft	13.756 m	Sea Condition	Rough	
Displacement	122069.3 ton	Wind Dir. & Velocity	S-20°, 10 m/s	
		Depth of Sea Water	129 m	
Trial Result (Rudder angle/Course= 10/10 deg)				
		Port Rudder First	Starboard Rudder First	
Clock Time		20 : 30	20 : 00	
Base Course		240°	240°	
Shaft rpm		100.0 rpm	100.0 rpm	
Ship speed		13.80 knot	13.80 knot	
Overshoot Angle	1st	16.9°	8.5°	
	2nd	27.5°	31.5°	
Trial Result (Rudder angle/Course= 20/20 deg)				
		Port Rudder First	Starboard Rudder First	
Clock Time		21 : 30	21 : 0105	
Base Course		240°	240°	
Shaft rpm		100.0 rpm	100.0 rpm	
Ship speed		13.80 knots	13.80 knots	
Overshoot Angle	1st	18.0°	19.6°	
	2nd	18.1°	19.2°	

"Z" MANEUVER

(BALLASTED CONDITION at IMO SPEED)

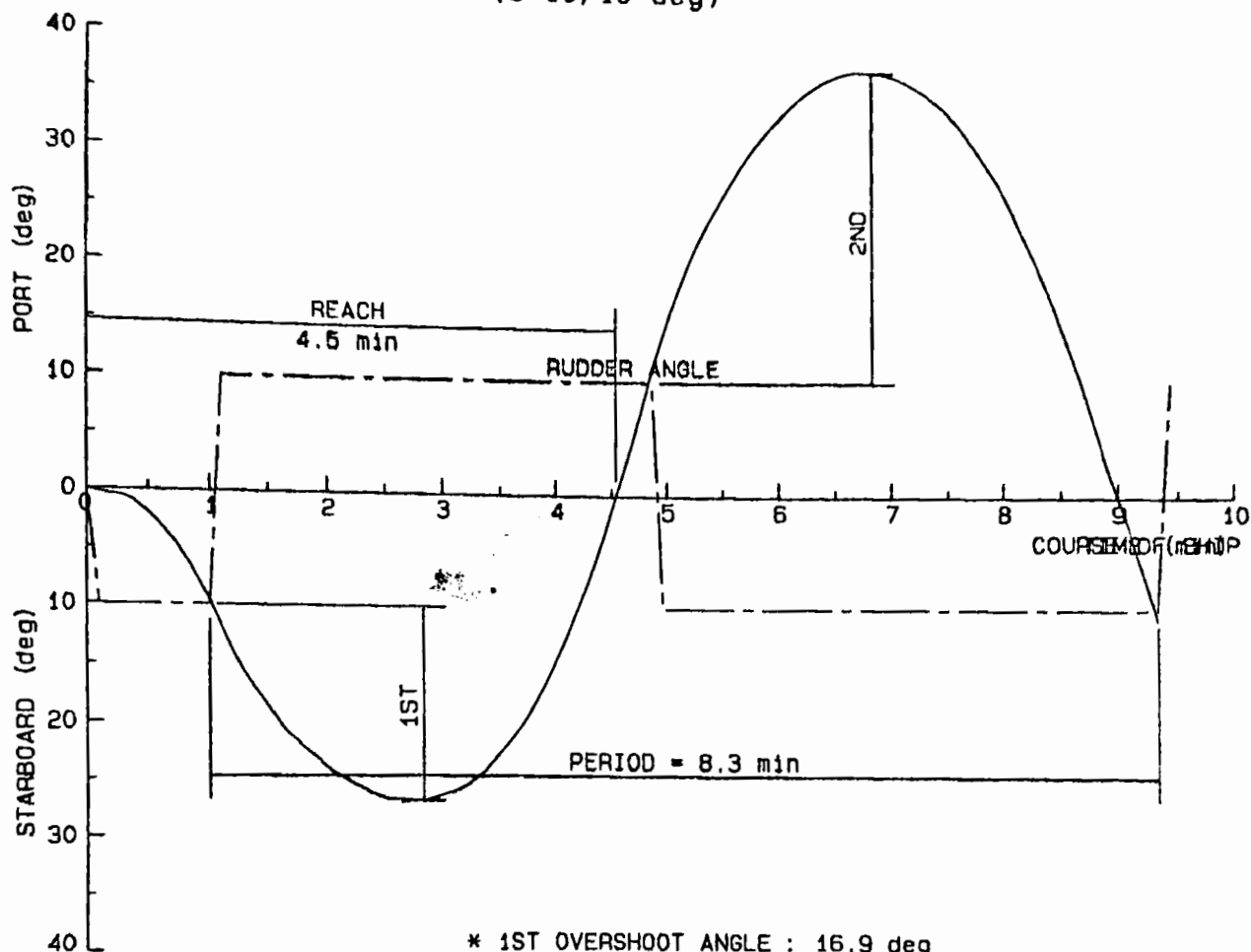
(P-10/10 deg)



"Z" MANEUVER

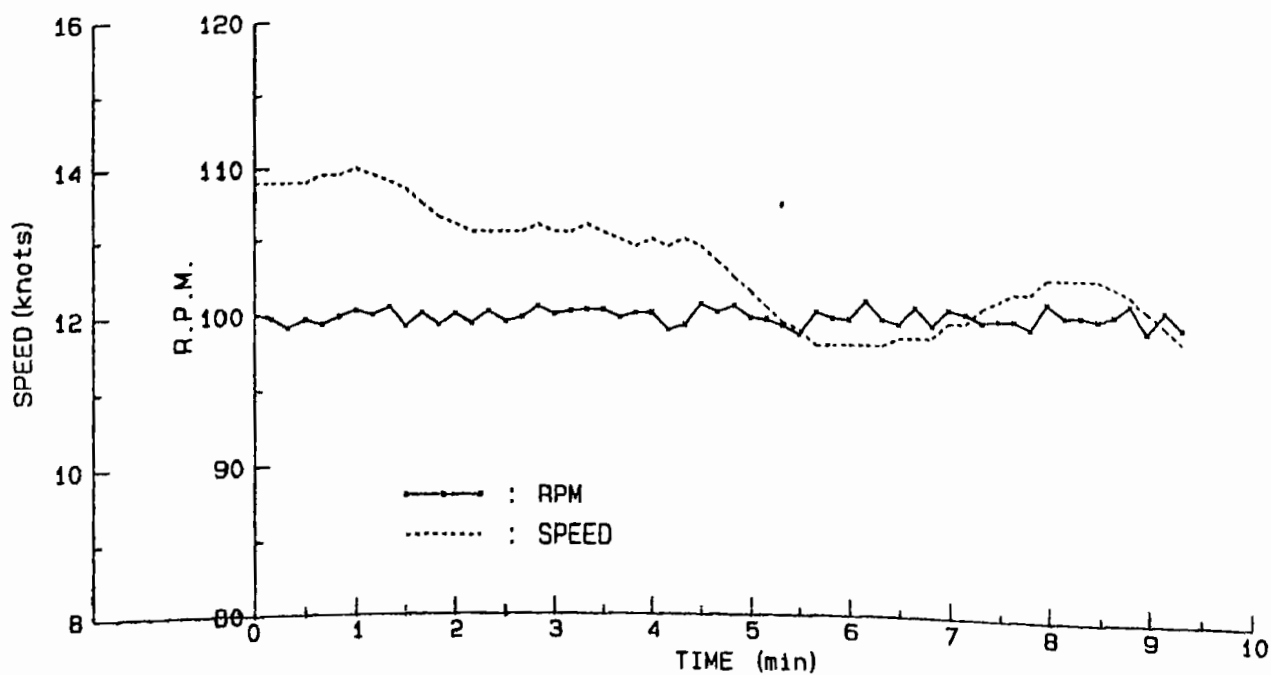
(BALLASTED CONDITION at IMO SPEED)

(S-10/10 deg)



* 1ST OVERSHOOT ANGLE : 16.9 deg

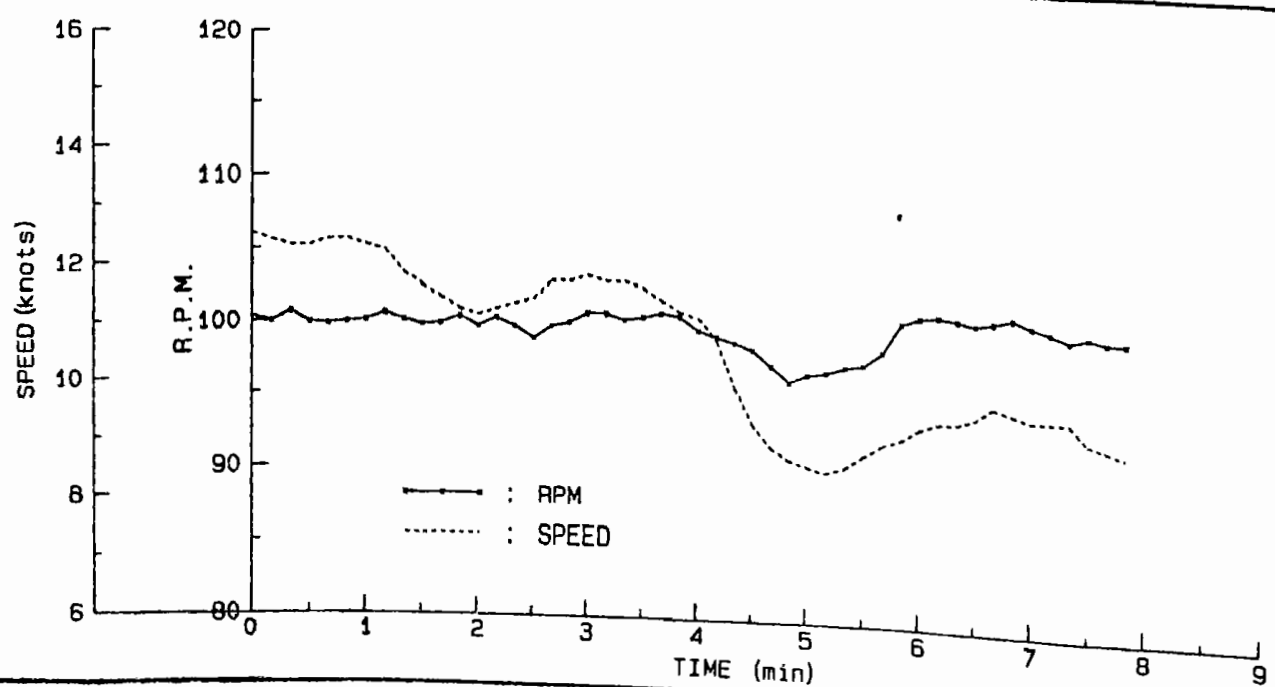
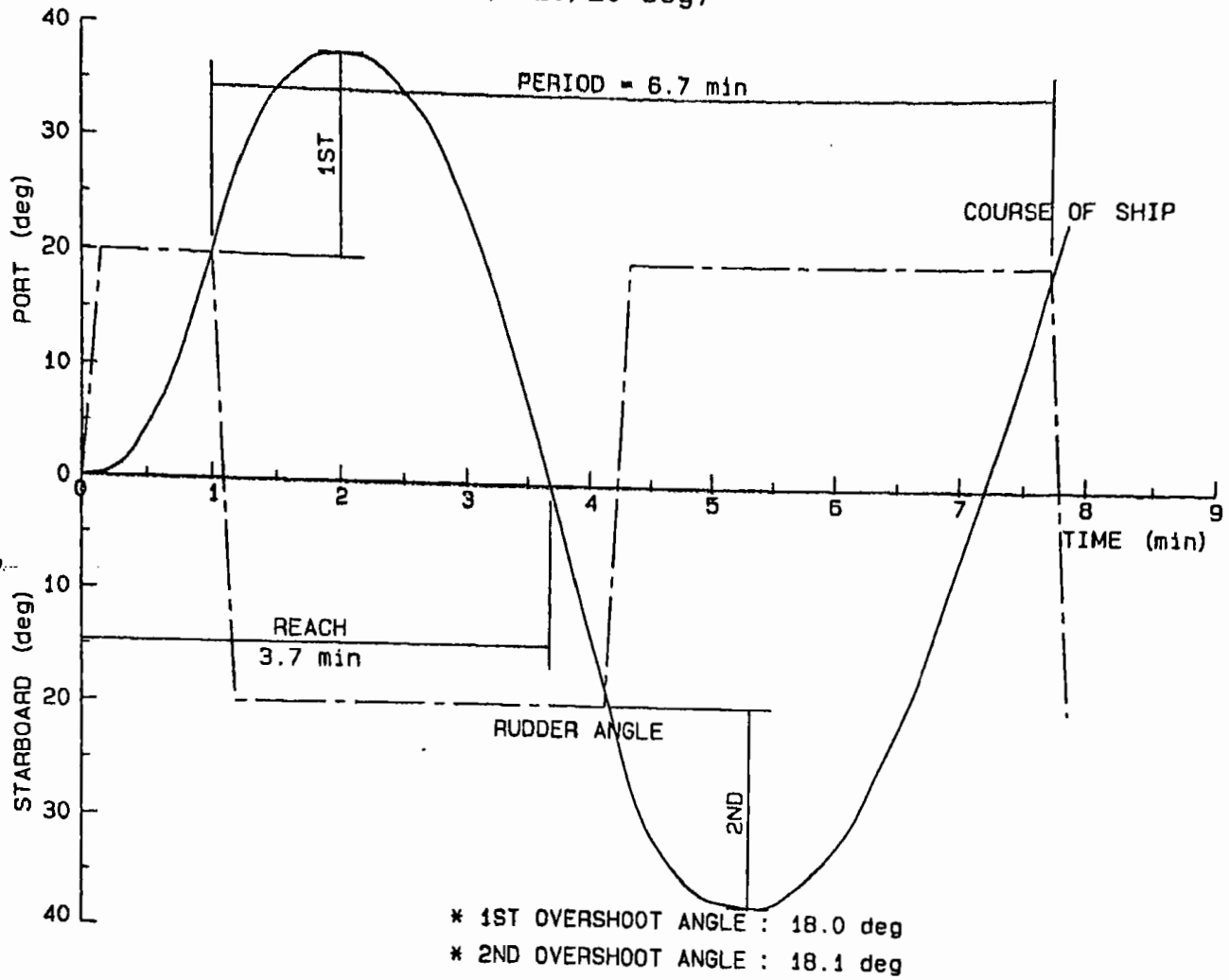
* 2ND OVERSHOOT ANGLE : 27.5 deg



"Z" MANEUVER

(BALLASTED CONDITION at IMO SPEED)

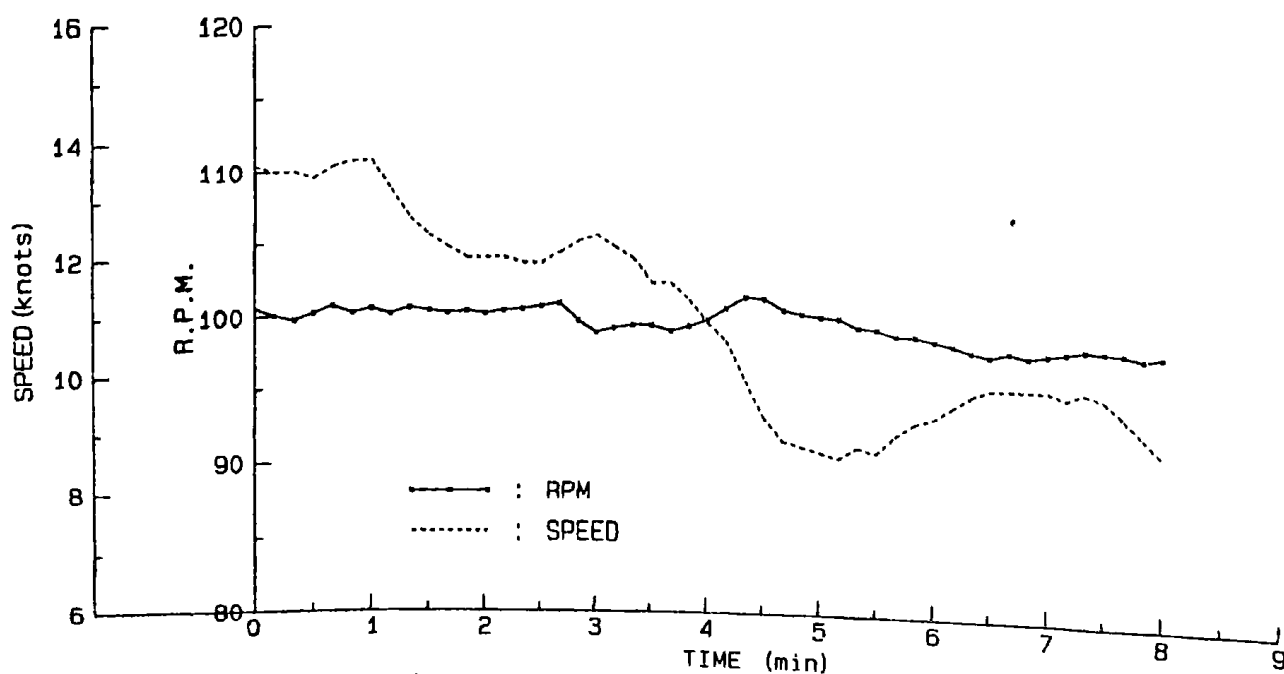
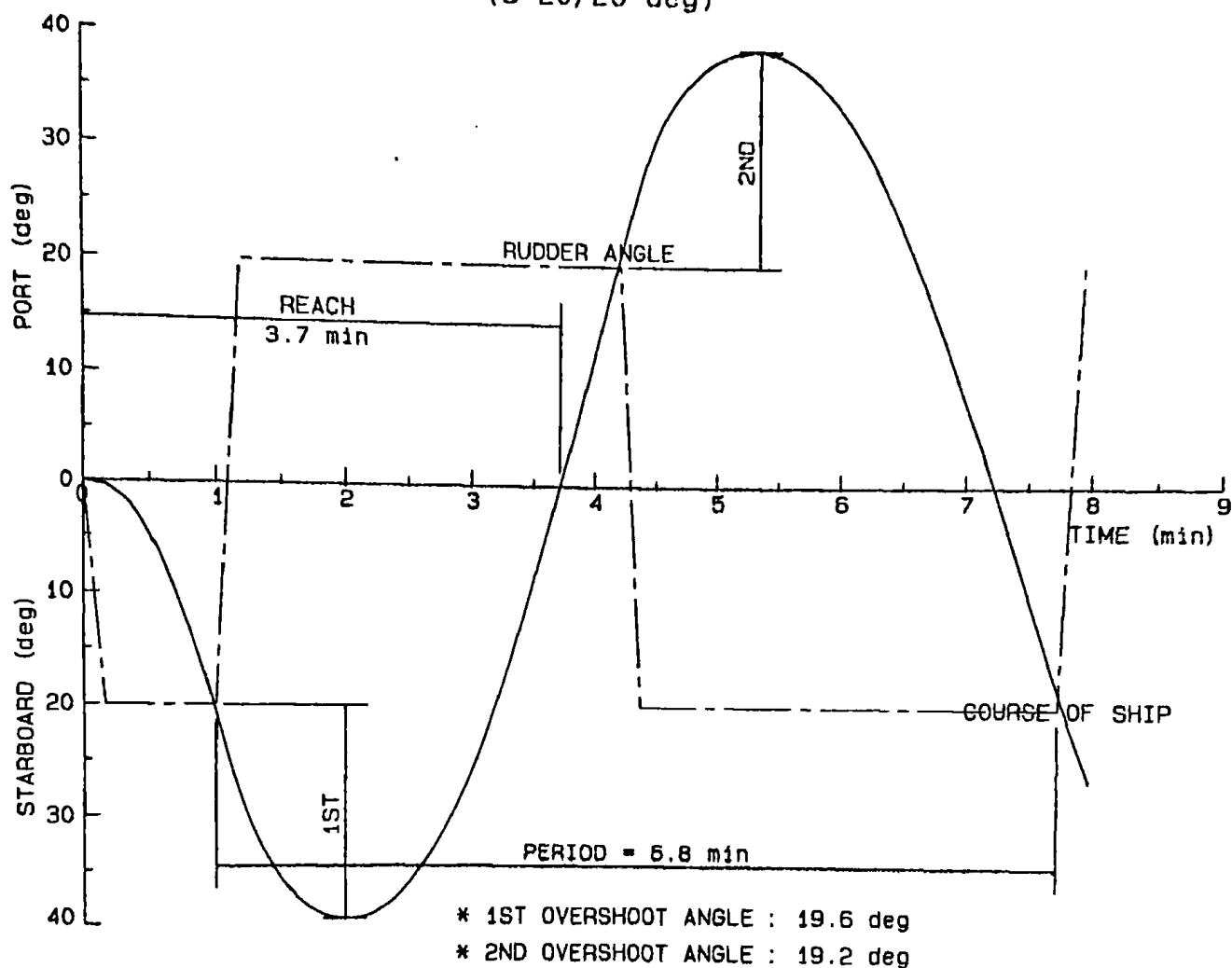
(P-20/20 deg)



"Z" MANEUVER

(BALLASTED CONDITION at IMO SPEED)

(S-20/20 deg)



INITIAL TURNING ABILITY

at 10/10 ZIGZAG TEST

(unit : meter)

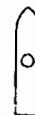
PORT

STARBOARD

Initial Speed
: 13.7 kts

Initial Speed
: 13.8 kts

* REMARK :



: 10 deg changed
position of course

-100 0 100

TEST RESULT

	PORT TURN	STB' D TURN
TRACK REACH	437.4 m (1.83 L)	435.2 m (1.82 L)
ELAPSED TIME	1 min 03 sec	1 min 07 sec

 HHIC	Test No. H-	Page	17
	Sea Trial Results for 6) Steering Gear Test	Ship No.	N208
		Date	'10.09.03

Trial date : 2010. 09. 03		Trial Place : OFF BUSAN	
Ship Condition : <i>Trial Ballast Condition</i>		Sea Condition :	
Draft, Fore	13.660 m	Weather	<i>Fine</i>
Draft, Aft	13.671 m	Sea Condition	<i>Calm</i>
Displacement	120847.2	Wind Dir. & Velocity	<i>P 20°, 15.3 m/s</i>
		Depth of Sea Water	<i>129 m</i>

Trial Result with No.1 Unit				
Clock Time	17 : 30			
Max. Rudder Angle	35°			
Shaft rpm	105 rpm			
Ship speed	16.4 knots			
Rudder Movement	Midship to P-30 deg	P-35 to S-30 deg	S-35 to P-30 deg	P-35 deg to Midship
Taken Time (sec)	11.56	25.83	25.8	14.27
Motor Ampere (A)	60 A	90 A	90 A	60 A
Max. Oil Pressure (bar)	140 bar	150 bar	180 bar	140 bar

Trial Result with No.2 Unit				
Clock Time	17 : 35			
Max. Rudder Angle	35°			
Shaft rpm	105 rpm			
Ship speed	15.10 knots			
Rudder Movement	Midship to -S30 deg	S-35 to P-30 deg	P-35 to S-30 deg	S-35 deg to Midship
Taken Time (sec)	11.66	25.21	26.45	13.84
Motor Ampere (A)	50 A	80 A	70 A	60 A
Max. Oil Pressure (bar)	100 bar	150 bar	150 bar	140 bar